SPATIOTEMPORAL FORMALIZATION OF BULLET-TIME MECHANICS

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Framework: Relativistic Asymmetric Game Loop Mechanics

1. Core Philosophy of Asymmetric Time Dilatation

In standard action mechanics, time scales uniformly across all system objects. Max Payne's Bullet-Time disrupts this by introducing asymmetric time dilatation (γ). This mechanism acts as a perceptual buffer, decoupling player cognitive input velocity from physics transformation vectors.

2. Mathematical Formulation of the Dilatation Curve

Let t_{global} represent the baseline simulation tick rate, and t_{player} represent the player camera/aim matrix transformation rate. Under operational state variables:

- **Active Bullet-Time:** $dt_{global} \rightarrow \gamma \cdot dt_{global}$, where $0 < \gamma \leq 0.20$.
- **Asymmetric Aim Window:** $dt_{player} \rightarrow \beta \cdot dt_{player}$, where $\gamma < \beta \leq 1.0$.

This asymmetric scaling ensures that while bullet and enemy transformations slow down drastically, the user retains localized agency to look and track vectors at near-normal speeds.

3. The Spatiotemporal Dive Matrix (The Third Axis)

The signature mechanic features a specialized translational velocity boost coupled with absolute temporal locks. When initiating a Shootdodge:

1. The spatial vector (X, Y) receives an instantaneous momentum impulse injection.
2. The Z-axis height modifier is driven by an irreversible gravitational animation arc curve.
3. Fire rates are normalized via an independent weapon tick loop, ensuring combat feedback remains punchy despite spatial time stretching.

4. Resource Depletion & Tactical Cooldown Loops

To balance operational states, a continuous linear resource drain function $R(t) = R_0 - \alpha \cdot \Delta t$ is evaluated. Regeneration occurs through active tactical mastery (neutralizing enemy entities), establishing a self-sustaining flow state.

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